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Part I

Cyclic analyses

1 crushing compression — solved tension tail — perfect bond

1.1 Cyclic — crushing/solved/perfect — mesh 30.5 mm — energy balance, uniaxial field — uniform grid — materials measured — isotropic $a_1 = a_3 = 0.02$ — top band 1000 mm — measured to $\pm 2.5\%$

2026-09-24_235853_cyclic_crushing-solved_perfect_guniform_m30.5_iso0.02_tb1000_d2.5pct_protomeasured

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	30.5 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	4,961 / 22,310
composition	4,961 nodes, 22,310 elements (19,360 concrete, 1,440 longitudinal, 1,510 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u / \varepsilon_{cr}$	orthogonal 38, diagonal 27.2
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,889.1 \text{ mm}^2$
$E_t A_t$ per strut	58.6e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8 \text{ MPa}$, $f_t = 2.03 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-47.6, epsc0:-0.003068, fpcu:-9.52, epsU:-0.024593, lambda:0.1, ft:2.03, Ets:837.916	1,052 concrete elements, L = 30.5 mm
2	Concrete02	fpc:-47.6, epsc0:-0.003068, fpcu:-9.52, epsU:-0.024544, lambda:0.1, ft:2.03, Ets:1184.99	960 concrete elements, L = 43.13 mm
3	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0266976, lambda:0.1, ft:2.03, Ets:837.916	8,708 concrete elements, L = 30.5 mm
4	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.022069, lambda:0.1, ft:2.03, Ets:1184.99	8,640 concrete elements, L = 43.13 mm
5	Steel02	Fy:434, E0:200000, b:0.02, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	1,440 longitudinal/stirrup elements, L = 30.5 mm
6	Steel02	Fy:448, E0:200000, b:0.02, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	1,510 longitudinal/stirrup elements, L = 30.5 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	1,052	corotTruss	30.5	1,889.1	31,030	5.862e7	1.922e6
concrete, orthogonal	8,708	corotTruss	30.5	1,889.1	31,030	5.862e7	1.922e6
concrete, diagonal	960	corotTruss	43.13	1,889.1	31,030	5.862e7	1.359e6
concrete, diagonal	8,640	corotTruss	43.13	1,889.1	31,030	5.862e7	1.359e6
longitudinal	960	corotTruss	30.5	142.7	200,000	2.853e7	9.355e5
longitudinal	480	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	480	corotTruss	30.5	35.6	200,000	7.118e6	2.334e5
stirrup	1,030	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5

Analysis and solver

analysis	cyclic
protocol	measured, 1 cycle(s) per level, largest $\pm 2.5\%$ drift
levels (% drift)	0.0833, 0.2022, 0.4372, 0.6653, 0.9003, 1.3716, 1.8361, 2.3224
drive path	1,144.6 mm
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 4,556 steps per fundamental period
first period T_1	20.105 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	34,128,580
wall clock	123.67 h
converged	True

Recorded response

peak base shear	139.9 kN (5 ms moving average)
raw sample maximum	142.7 kN
drift at peak	0.437%
drift capacity (20% drop)	not reached — basis: cyclic tip envelope, 16 tips over 12 levels
traced to	0% drift
inertia + damping in the reaction, ascending branch (p95)	1.3 kN

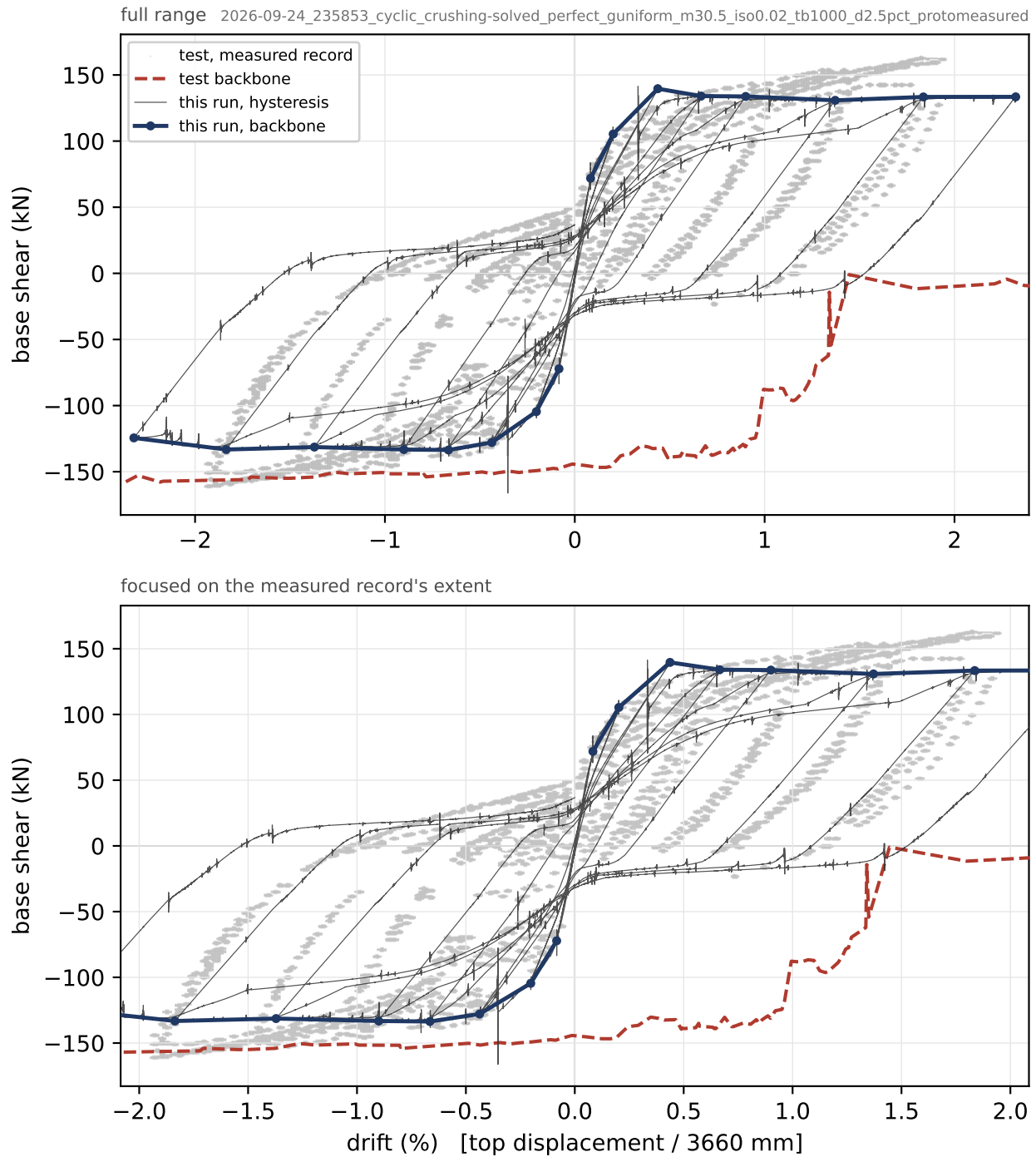


Figure 1: Hysteresis and its backbone over the measured record and its envelope; full range above, the record's own extent below.

Part II

Pushover analyses

2 crushing compression — solved tension tail — perfect bond

2.1 Pushover — crushing/solved/perfect — mesh 25 mm — energy balance, uniaxial field — to 0.01% drift

2026-09-20_173801_pushover_crushing-solved_perfect_d0.01pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	25 mm
strut horizon	$1.5 \times$ mesh
nodes / elements	7,446 / 32,423
composition	7,446 nodes, 32,423 elements (29,195 concrete, 1,740 longitudinal, 1,488 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 45.3, diagonal 32.4
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,560.6 \text{ mm}^2$
$E_t A_t$ per strut	48.4e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8$ MPa, $f_t = 2.03$ MPa, $G_f = 0.075$ N/mm
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414$ MPa, $E_s = 200,000$ MPa, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	materials, steel_iso, top_band, top_band_pitch — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0411871, lambda:0.1, ft:2.03, Ets:521.981	394 concrete elements, L = 19 mm
2	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0299316, lambda:0.1, ft:2.03, Ets:738.192	8 concrete elements, L = 26.87 mm
3	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0313916, lambda:0.1, ft:2.03, Ets:700.553	1,752 concrete elements, L = 25.5 mm
4	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0257189, lambda:0.1, ft:2.03, Ets:873.635	48 concrete elements, L = 31.8 mm
5	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0316751, lambda:0.1, ft:2.03, Ets:693.685	1,752 concrete elements, L = 25.25 mm
6	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0258643, lambda:0.1, ft:2.03, Ets:868.137	48 concrete elements, L = 31.6 mm
7	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0333404, lambda:0.1, ft:2.03, Ets:655.91	3,504 concrete elements, L = 23.88 mm
8	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0266878, lambda:0.1, ft:2.03, Ets:838.261	96 concrete elements, L = 30.51 mm
9	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0315799, lambda:0.1, ft:2.03, Ets:695.974	7,191 concrete elements, L = 25.33 mm
10	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0258157, lambda:0.1, ft:2.03, Ets:869.968	564 concrete elements, L = 31.67 mm
11	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0319642, lambda:0.1, ft:2.03, Ets:686.817	102 concrete elements, L = 25 mm
12	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.026011, lambda:0.1, ft:2.03, Ets:862.659	8 concrete elements, L = 31.4 mm
13	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0232046, lambda:0.1, ft:2.03, Ets:981.067	48 concrete elements, L = 35.71 mm
14	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0233071, lambda:0.1, ft:2.03, Ets:976.174	48 concrete elements, L = 35.53 mm
15	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0238799, lambda:0.1, ft:2.03, Ets:949.703	96 concrete elements, L = 34.57 mm
16	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0230714, lambda:0.1, ft:2.03, Ets:987.499	3,384 concrete elements, L = 35.94 mm

tag	type	arguments	applied to
17	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0231719, lambda:0.1, ft:2.03, Ets:982.639	3,384 concrete ele- ments, L = 35.77 mm
18	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0237331, lambda:0.1, ft:2.03, Ets:956.346	6,768 concrete ele- ments, L = 34.81 mm
19	Steel02	Fy:414, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	3,228 longitudinal/stirrup elements

Strut families — axial stiffness

note E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete	8	corotTruss	19	1,186.1	31,030	3.680e7	1.937e6
concrete	4	corotTruss	19	1,388.9	31,030	4.310e7	2.268e6
concrete	20	corotTruss	19	1,591.8	31,030	4.939e7	2.600e6
concrete	4	corotTruss	19	1,584	31,030	4.915e7	2.587e6
concrete	20	corotTruss	19	1,576.2	31,030	4.891e7	2.574e6
concrete	4	corotTruss	19	1,533.3	31,030	4.758e7	2.504e6
concrete	46	corotTruss	19	1,490.4	31,030	4.625e7	2.434e6
concrete	2	corotTruss	19	1,383.7	31,030	4.294e7	2.260e6
concrete	188	corotTruss	19	1,581.4	31,030	4.907e7	2.583e6
concrete	92	corotTruss	19	1,581.4	31,030	4.907e7	2.583e6
concrete	2	corotTruss	19	1,571	31,030	4.875e7	2.566e6
concrete	4	corotTruss	19	1,560.6	31,030	4.843e7	2.549e6
concrete	48	corotTruss	23.88	1,186.1	31,030	3.680e7	1.541e6
concrete	48	corotTruss	23.88	1,560.6	31,030	4.843e7	2.028e6
concrete	24	corotTruss	23.88	1,383.7	31,030	4.294e7	1.798e6
concrete	2,256	corotTruss	23.88	1,581.4	31,030	4.907e7	2.055e6
concrete	1,104	corotTruss	23.88	1,581.4	31,030	4.907e7	2.055e6
concrete	24	corotTruss	23.88	1,571	31,030	4.875e7	2.041e6
concrete, orthogonal	4	corotTruss	25	1,186.1	31,030	3.680e7	1.472e6
concrete, orthogonal	4	corotTruss	25	1,388.9	31,030	4.310e7	1.724e6
concrete, orthogonal	20	corotTruss	25	1,591.8	31,030	4.939e7	1.976e6
concrete, orthogonal	4	corotTruss	25	1,584	31,030	4.915e7	1.966e6
concrete, orthogonal	20	corotTruss	25	1,576.2	31,030	4.891e7	1.956e6
concrete, orthogonal	4	corotTruss	25	1,533.3	31,030	4.758e7	1.903e6
concrete, orthogonal	46	corotTruss	25	1,490.4	31,030	4.625e7	1.850e6
concrete	24	corotTruss	25.25	1,186.1	31,030	3.680e7	1.458e6
concrete	24	corotTruss	25.25	1,560.6	31,030	4.843e7	1.918e6
concrete	12	corotTruss	25.25	1,383.7	31,030	4.294e7	1.700e6
concrete	1,128	corotTruss	25.25	1,581.4	31,030	4.907e7	1.943e6
concrete	552	corotTruss	25.25	1,581.4	31,030	4.907e7	1.943e6
concrete	12	corotTruss	25.25	1,571	31,030	4.875e7	1.931e6
concrete	282	corotTruss	25.33	1,186.1	31,030	3.680e7	1.453e6
concrete	282	corotTruss	25.33	1,388.9	31,030	4.310e7	1.701e6
concrete	1,410	corotTruss	25.33	1,591.8	31,030	4.939e7	1.950e6
concrete	282	corotTruss	25.33	1,584	31,030	4.915e7	1.940e6
concrete	1,410	corotTruss	25.33	1,576.2	31,030	4.891e7	1.931e6
concrete	282	corotTruss	25.33	1,533.3	31,030	4.758e7	1.878e6
concrete	3,243	corotTruss	25.33	1,490.4	31,030	4.625e7	1.826e6

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete	24	corotTruss	25.5	1,186.1	31,030	3.680e7	1.443e6
concrete	24	corotTruss	25.5	1,560.6	31,030	4.843e7	1.899e6
concrete	12	corotTruss	25.5	1,383.7	31,030	4.294e7	1.684e6
concrete	1,128	corotTruss	25.5	1,581.4	31,030	4.907e7	1.924e6
concrete	552	corotTruss	25.5	1,581.4	31,030	4.907e7	1.924e6
concrete	12	corotTruss	25.5	1,571	31,030	4.875e7	1.912e6
concrete	8	corotTruss	26.87	1,186.1	31,030	3.680e7	1.370e6
concrete	96	corotTruss	30.51	1,329.5	31,030	4.126e7	1.352e6
concrete	8	corotTruss	31.4	1,360.5	31,030	4.222e7	1.344e6
concrete	48	corotTruss	31.6	1,367.3	31,030	4.243e7	1.343e6
concrete	564	corotTruss	31.67	1,369.5	31,030	4.250e7	1.342e6
concrete	48	corotTruss	31.8	1,374	31,030	4.264e7	1.341e6
concrete	96	corotTruss	34.57	1,525.1	31,030	4.732e7	1.369e6
concrete	6,768	corotTruss	34.81	1,535.2	31,030	4.764e7	1.369e6
concrete, diagonal	48	corotTruss	35.53	1,568.4	31,030	4.867e7	1.370e6
concrete	48	corotTruss	35.71	1,576.1	31,030	4.891e7	1.370e6
concrete	3,384	corotTruss	35.77	1,578.8	31,030	4.899e7	1.370e6
concrete	2,256	corotTruss	35.94	1,586.6	31,030	4.923e7	1.370e6
concrete	1,128	corotTruss	35.94	1,586.6	31,030	4.923e7	1.370e6
longitudinal	16	corotTruss	19	142.7	200,000	2.853e7	1.502e6
longitudinal	8	corotTruss	19	63.3	200,000	1.267e7	6.667e5
longitudinal	16	corotTruss	25	142.7	200,000	2.853e7	1.141e6
longitudinal	8	corotTruss	25	63.3	200,000	1.267e7	5.067e5
longitudinal	1,128	corotTruss	25.33	142.7	200,000	2.853e7	1.126e6
longitudinal	564	corotTruss	25.33	63.3	200,000	1.267e7	5.001e5
stirrup	456	corotTruss	23.88	63.3	200,000	1.267e7	5.305e5
stirrup	228	corotTruss	25.25	63.3	200,000	1.267e7	5.017e5
stirrup	576	corotTruss	25.5	35.6	200,000	7.118e6	2.791e5
stirrup	228	corotTruss	25.5	63.3	200,000	1.267e7	4.968e5

Analysis and solver

analysis	pushover
target	0.01% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	2.739 us, 7,305 steps per fundamental period
first period T_1	20.009 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	17,583
wall clock	0.1 h
converged	True

Recorded response

peak base shear	9.8 kN (5 ms moving average)
raw sample maximum	10.4 kN
drift at peak	0.009%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	0.01% drift
inertia + damping in the reaction,	2.2 kN
ascending branch (p95)	

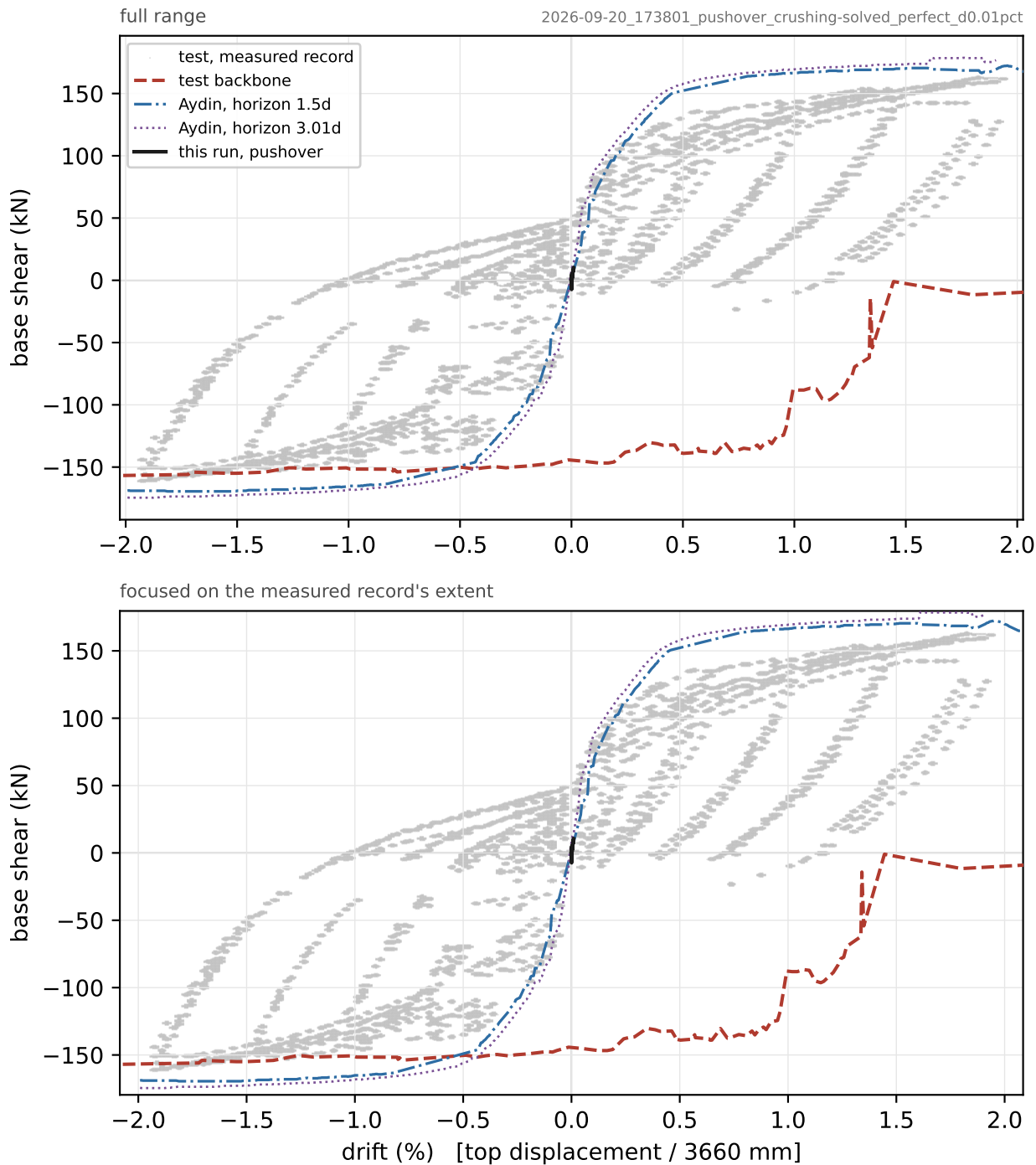


Figure 2: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

2.2 Pushover — crushing/solved/perfect — mesh 25 mm — energy balance, uniaxial field — to 2.5% drift

2026-09-20_174435_pushover_crushing-solved_perfect_d2.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	25 mm
strut horizon	$1.5 \times$ mesh
nodes / elements	7,446 / 32,423
composition	7,446 nodes, 32,423 elements (29,195 concrete, 1,740 longitudinal, 1,488 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 45.3, diagonal 32.4
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,560.6 \text{ mm}^2$
$E_t A_t$ per strut	48.4e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8$ MPa, $f_t = 2.03$ MPa, $G_f = 0.075$ N/mm
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414$ MPa, $E_s = 200,000$ MPa, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	materials, steel_iso, top_band, top_band_pitch — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0411871, lambda:0.1, ft:2.03, Ets:521.981	394 concrete elements, L = 19 mm
2	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0299316, lambda:0.1, ft:2.03, Ets:738.192	8 concrete elements, L = 26.87 mm
3	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0313916, lambda:0.1, ft:2.03, Ets:700.553	1,752 concrete elements, L = 25.5 mm
4	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0257189, lambda:0.1, ft:2.03, Ets:873.635	48 concrete elements, L = 31.8 mm
5	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0316751, lambda:0.1, ft:2.03, Ets:693.685	1,752 concrete elements, L = 25.25 mm
6	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0258643, lambda:0.1, ft:2.03, Ets:868.137	48 concrete elements, L = 31.6 mm
7	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0333404, lambda:0.1, ft:2.03, Ets:655.91	3,504 concrete elements, L = 23.88 mm
8	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0266878, lambda:0.1, ft:2.03, Ets:838.261	96 concrete elements, L = 30.51 mm
9	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0315799, lambda:0.1, ft:2.03, Ets:695.974	7,191 concrete elements, L = 25.33 mm
10	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0258157, lambda:0.1, ft:2.03, Ets:869.968	564 concrete elements, L = 31.67 mm
11	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0319642, lambda:0.1, ft:2.03, Ets:686.817	102 concrete elements, L = 25 mm
12	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.026011, lambda:0.1, ft:2.03, Ets:862.659	8 concrete elements, L = 31.4 mm
13	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0232046, lambda:0.1, ft:2.03, Ets:981.067	48 concrete elements, L = 35.71 mm
14	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0233071, lambda:0.1, ft:2.03, Ets:976.174	48 concrete elements, L = 35.53 mm
15	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0238799, lambda:0.1, ft:2.03, Ets:949.703	96 concrete elements, L = 34.57 mm
16	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0230714, lambda:0.1, ft:2.03, Ets:987.499	3,384 concrete elements, L = 35.94 mm

tag	type	arguments	applied to
17	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0231719, lambda:0.1, ft:2.03, Ets:982.639	3,384 concrete ele- ments, L = 35.77 mm
18	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0237331, lambda:0.1, ft:2.03, Ets:956.346	6,768 concrete ele- ments, L = 34.81 mm
19	Steel02	Fy:414, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	3,228 longitudinal/stirrup elements

Strut families — axial stiffness

note E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete	8	corotTruss	19	1,186.1	31,030	3.680e7	1.937e6
concrete	4	corotTruss	19	1,388.9	31,030	4.310e7	2.268e6
concrete	20	corotTruss	19	1,591.8	31,030	4.939e7	2.600e6
concrete	4	corotTruss	19	1,584	31,030	4.915e7	2.587e6
concrete	20	corotTruss	19	1,576.2	31,030	4.891e7	2.574e6
concrete	4	corotTruss	19	1,533.3	31,030	4.758e7	2.504e6
concrete	46	corotTruss	19	1,490.4	31,030	4.625e7	2.434e6
concrete	2	corotTruss	19	1,383.7	31,030	4.294e7	2.260e6
concrete	188	corotTruss	19	1,581.4	31,030	4.907e7	2.583e6
concrete	92	corotTruss	19	1,581.4	31,030	4.907e7	2.583e6
concrete	2	corotTruss	19	1,571	31,030	4.875e7	2.566e6
concrete	4	corotTruss	19	1,560.6	31,030	4.843e7	2.549e6
concrete	48	corotTruss	23.88	1,186.1	31,030	3.680e7	1.541e6
concrete	48	corotTruss	23.88	1,560.6	31,030	4.843e7	2.028e6
concrete	24	corotTruss	23.88	1,383.7	31,030	4.294e7	1.798e6
concrete	2,256	corotTruss	23.88	1,581.4	31,030	4.907e7	2.055e6
concrete	1,104	corotTruss	23.88	1,581.4	31,030	4.907e7	2.055e6
concrete	24	corotTruss	23.88	1,571	31,030	4.875e7	2.041e6
concrete, orthogonal	4	corotTruss	25	1,186.1	31,030	3.680e7	1.472e6
concrete, orthogonal	4	corotTruss	25	1,388.9	31,030	4.310e7	1.724e6
concrete, orthogonal	20	corotTruss	25	1,591.8	31,030	4.939e7	1.976e6
concrete, orthogonal	4	corotTruss	25	1,584	31,030	4.915e7	1.966e6
concrete, orthogonal	20	corotTruss	25	1,576.2	31,030	4.891e7	1.956e6
concrete, orthogonal	4	corotTruss	25	1,533.3	31,030	4.758e7	1.903e6
concrete, orthogonal	46	corotTruss	25	1,490.4	31,030	4.625e7	1.850e6
concrete	24	corotTruss	25.25	1,186.1	31,030	3.680e7	1.458e6
concrete	24	corotTruss	25.25	1,560.6	31,030	4.843e7	1.918e6
concrete	12	corotTruss	25.25	1,383.7	31,030	4.294e7	1.700e6
concrete	1,128	corotTruss	25.25	1,581.4	31,030	4.907e7	1.943e6
concrete	552	corotTruss	25.25	1,581.4	31,030	4.907e7	1.943e6
concrete	12	corotTruss	25.25	1,571	31,030	4.875e7	1.931e6
concrete	282	corotTruss	25.33	1,186.1	31,030	3.680e7	1.453e6
concrete	282	corotTruss	25.33	1,388.9	31,030	4.310e7	1.701e6
concrete	1,410	corotTruss	25.33	1,591.8	31,030	4.939e7	1.950e6
concrete	282	corotTruss	25.33	1,584	31,030	4.915e7	1.940e6
concrete	1,410	corotTruss	25.33	1,576.2	31,030	4.891e7	1.931e6
concrete	282	corotTruss	25.33	1,533.3	31,030	4.758e7	1.878e6
concrete	3,243	corotTruss	25.33	1,490.4	31,030	4.625e7	1.826e6

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete	24	corotTruss	25.5	1,186.1	31,030	3.680e7	1.443e6
concrete	24	corotTruss	25.5	1,560.6	31,030	4.843e7	1.899e6
concrete	12	corotTruss	25.5	1,383.7	31,030	4.294e7	1.684e6
concrete	1,128	corotTruss	25.5	1,581.4	31,030	4.907e7	1.924e6
concrete	552	corotTruss	25.5	1,581.4	31,030	4.907e7	1.924e6
concrete	12	corotTruss	25.5	1,571	31,030	4.875e7	1.912e6
concrete	8	corotTruss	26.87	1,186.1	31,030	3.680e7	1.370e6
concrete	96	corotTruss	30.51	1,329.5	31,030	4.126e7	1.352e6
concrete	8	corotTruss	31.4	1,360.5	31,030	4.222e7	1.344e6
concrete	48	corotTruss	31.6	1,367.3	31,030	4.243e7	1.343e6
concrete	564	corotTruss	31.67	1,369.5	31,030	4.250e7	1.342e6
concrete	48	corotTruss	31.8	1,374	31,030	4.264e7	1.341e6
concrete	96	corotTruss	34.57	1,525.1	31,030	4.732e7	1.369e6
concrete	6,768	corotTruss	34.81	1,535.2	31,030	4.764e7	1.369e6
concrete, diagonal	48	corotTruss	35.53	1,568.4	31,030	4.867e7	1.370e6
concrete	48	corotTruss	35.71	1,576.1	31,030	4.891e7	1.370e6
concrete	3,384	corotTruss	35.77	1,578.8	31,030	4.899e7	1.370e6
concrete	2,256	corotTruss	35.94	1,586.6	31,030	4.923e7	1.370e6
concrete	1,128	corotTruss	35.94	1,586.6	31,030	4.923e7	1.370e6
longitudinal	16	corotTruss	19	142.7	200,000	2.853e7	1.502e6
longitudinal	8	corotTruss	19	63.3	200,000	1.267e7	6.667e5
longitudinal	16	corotTruss	25	142.7	200,000	2.853e7	1.141e6
longitudinal	8	corotTruss	25	63.3	200,000	1.267e7	5.067e5
longitudinal	1,128	corotTruss	25.33	142.7	200,000	2.853e7	1.126e6
longitudinal	564	corotTruss	25.33	63.3	200,000	1.267e7	5.001e5
stirrup	456	corotTruss	23.88	63.3	200,000	1.267e7	5.305e5
stirrup	228	corotTruss	25.25	63.3	200,000	1.267e7	5.017e5
stirrup	576	corotTruss	25.5	35.6	200,000	7.118e6	2.791e5
stirrup	228	corotTruss	25.5	63.3	200,000	1.267e7	4.968e5

Analysis and solver

analysis	pushover
target	2.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	2.739 us, 7,305 steps per fundamental period
first period T_1	20.009 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	4,395,484
wall clock	33.19 h
converged	True

Recorded response

peak base shear	138.3 kN (5 ms moving average)
raw sample maximum	145 kN
drift at peak	0.519%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	2.5% drift
inertia + damping in the reaction, ascending branch (p95)	0.9 kN

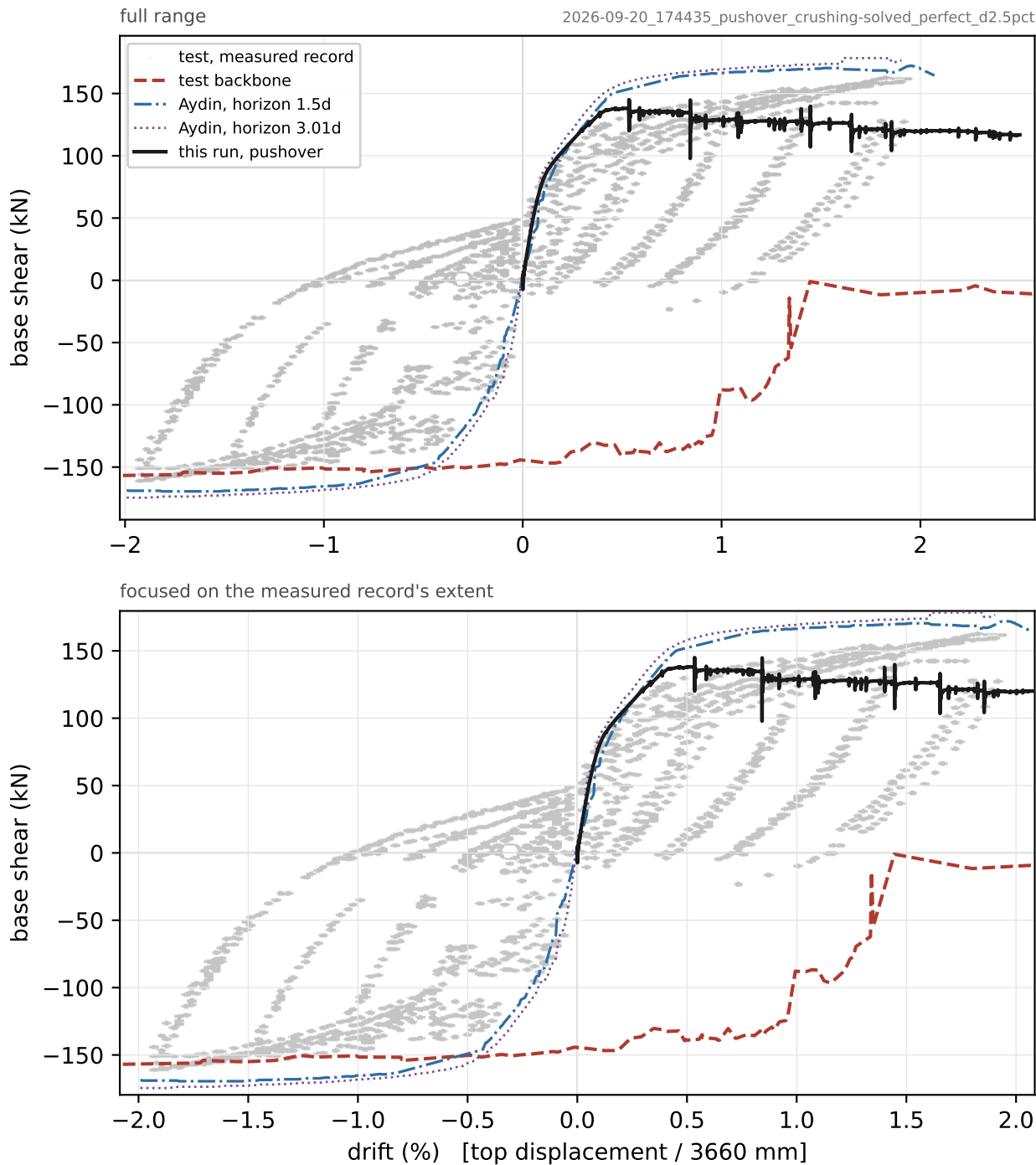


Figure 3: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

2.3 Pushover — crushing/solved/perfect — mesh 30.5 mm — energy balance, uniaxial field — uniform grid — $b = 0$ — $f_y = 454$ — to 1.5% drift

2026-09-21_085208_pushover_crushing-solved_perfect_uniform_m30.5_sb0_fy454_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	30.5 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	4,961 / 22,002
composition	4,961 nodes, 22,002 elements (19,360 concrete, 1,440 longitudinal, 1,202 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 38, diagonal 27.2
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,889.1 \text{ mm}^2$
$E_t A_t$ per strut	58.6 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8 \text{ MPa}$, $f_t = 2.03 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	materials, steel_iso, top_band, top_band_pitch — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0266976, lambda:0.1, ft:2.03, Ets:837.916	9,760 concrete elements, L = 30.5 mm
2	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.022069, lambda:0.1, ft:2.03, Ets:1184.99	9,600 concrete elements, L = 43.13 mm
3	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
4	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
5	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
6	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
7	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
8	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
9	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
10	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
11	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
12	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
13	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
14	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
15	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
16	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
17	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm

[illegible]

[illegible]

[illegible]

tag	type	arguments	applied to
105	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
106	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
107	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
108	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
109	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
110	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
111	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
112	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
113	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
114	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
115	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
116	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
117	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
118	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
119	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
120	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
121	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
122	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
123	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
124	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
125	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
126	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
127	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
128	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm
129	Steel02	Fy:454, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	38 stirrup elements, L = 30.5 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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[illegible]

[illegible]

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	38	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 4,556 steps per fundamental period
first period T_1	20.12 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,635,767
wall clock	10.86 h
converged	True

Recorded response

peak base shear	145.3 kN (5 ms moving average)
raw sample maximum	158.2 kN
drift at peak	0.5%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	1.3 kN

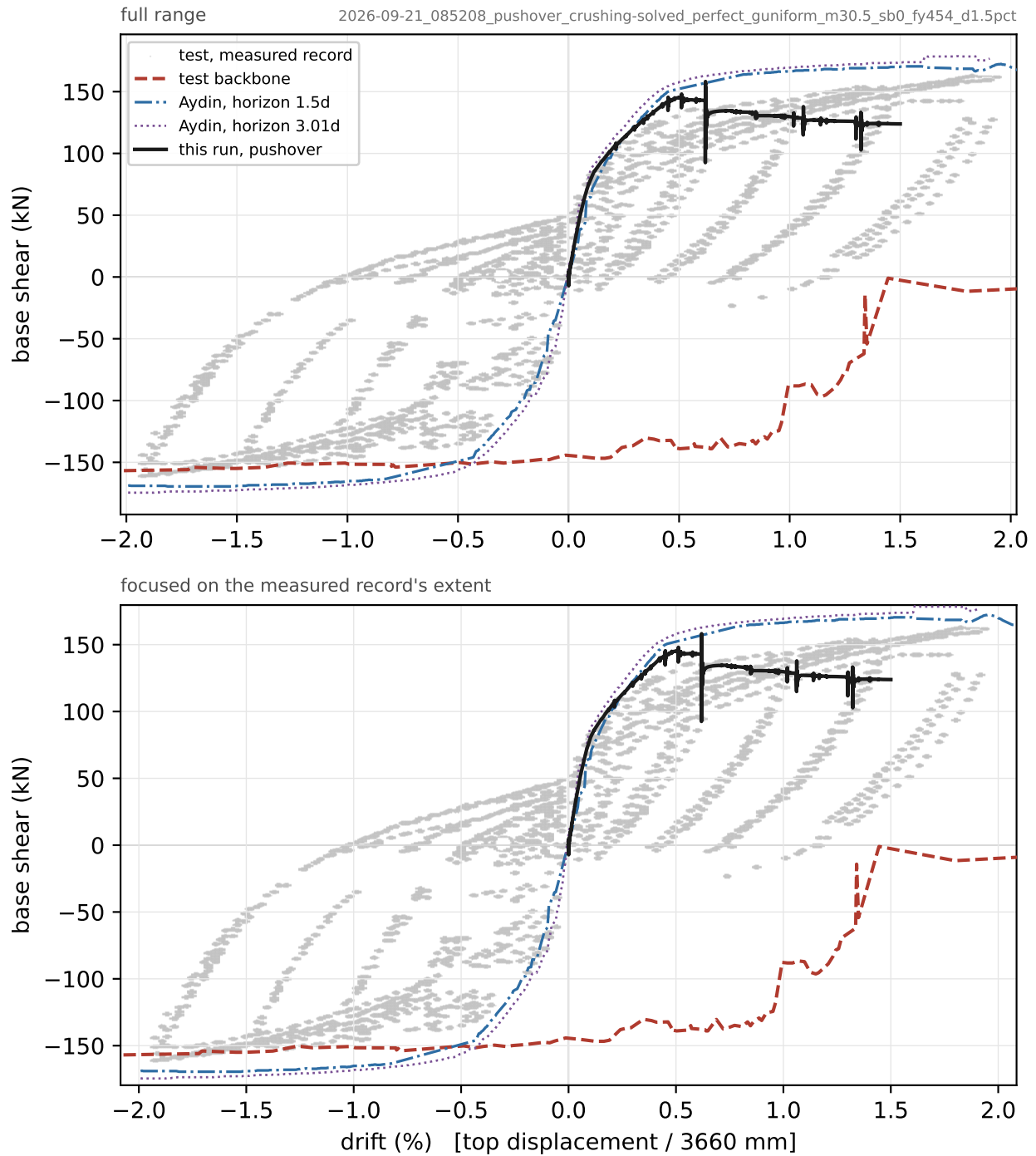


Figure 4: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

2.4 Pushover — crushing/solved/perfect — mesh 30.5 mm — energy balance, uniaxial field — uniform grid — $f_y = 454$ — to 1.5% drift

2026-09-21_085207_pushover_crushing-solved_perfect_guniform_m30.5_fy454_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	30.5 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	4,961 / 22,002
composition	4,961 nodes, 22,002 elements (19,360 concrete, 1,440 longitudinal, 1,202 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 38, diagonal 27.2
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,889.1 \text{ mm}^2$
$E_t A_t$ per strut	58.6 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8 \text{ MPa}$, $f_t = 2.03 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	materials, steel_iso, top_band, top_band_pitch — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0266976, lambda:0.1, ft:2.03, Ets:837.916	9,760 concrete elements, L = 30.5 mm
2	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.022069, lambda:0.1, ft:2.03, Ets:1184.99	9,600 concrete elements, L = 43.13 mm
3	Steel02	Fy:454, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	2,642 longitudinal/stirrup elements, L = 30.5 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	9,760	corotTruss	30.5	1,889.1	31,030	5.862e7	1.922e6
concrete, diagonal	9,600	corotTruss	43.13	1,889.1	31,030	5.862e7	1.359e6
longitudinal	960	corotTruss	30.5	142.7	200,000	2.853e7	9.355e5
longitudinal	480	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	480	corotTruss	30.5	35.6	200,000	7.118e6	2.334e5
stirrup	722	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 4,556 steps per fundamental period
first period T_1	20.12 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,635,767
wall clock	10.84 h
converged	True

Recorded response

peak base shear	145.7 kN (5 ms moving average)
raw sample maximum	161.4 kN
drift at peak	0.512%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction,	1.3 kN
ascending branch (p95)	

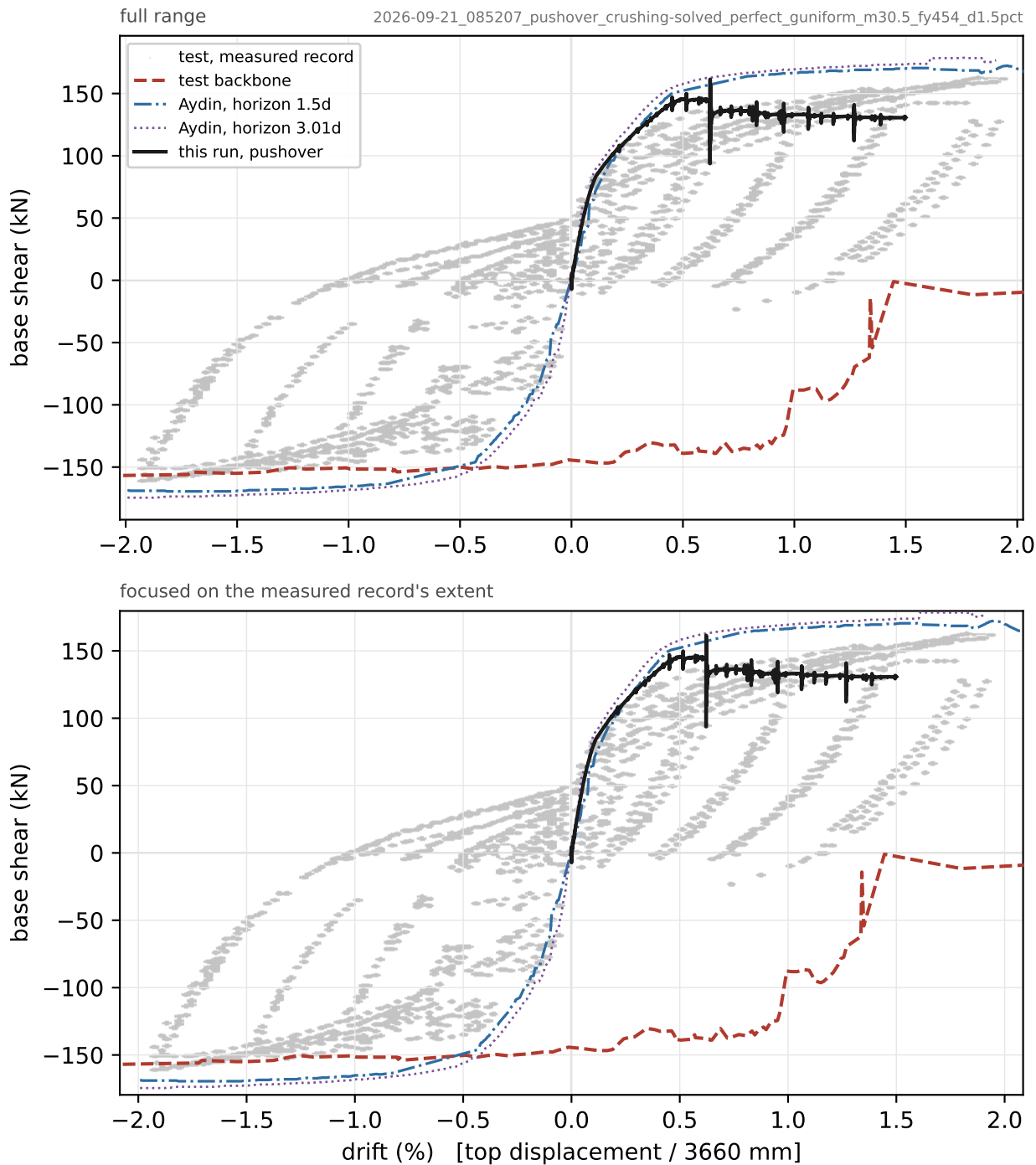


Figure 5: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

2.5 Pushover — crushing/solved/perfect — mesh 30.5 mm — energy balance, uniaxial field — uniform grid — to 2.5% drift

2026-09-20_174437_pushover_crushing-solved_perfect_guniform_m30.5_d2.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	30.5 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	4,961 / 22,002
composition	4,961 nodes, 22,002 elements (19,360 concrete, 1,440 longitudinal, 1,202 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 38, diagonal 27.2
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,889.1 \text{ mm}^2$
$E_t A_t$ per strut	58.6 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8 \text{ MPa}$, $f_t = 2.03 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	materials, steel_iso, top_band, top_band_pitch — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.0266976, lambda:0.1, ft:2.03, Ets:837.916	9,760 concrete elements, L = 30.5 mm
2	Concrete02	fpc:-42.8, epsc0:-0.00275862, fpcu:-8.56, epsU:-0.022069, lambda:0.1, ft:2.03, Ets:1184.99	9,600 concrete elements, L = 43.13 mm
3	Steel02	Fy:414, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	2,642 longitudinal/stirrup elements, L = 30.5 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	9,760	corotTruss	30.5	1,889.1	31,030	5.862e7	1.922e6
concrete, diagonal	9,600	corotTruss	43.13	1,889.1	31,030	5.862e7	1.359e6
longitudinal	960	corotTruss	30.5	142.7	200,000	2.853e7	9.355e5
longitudinal	480	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	480	corotTruss	30.5	35.6	200,000	7.118e6	2.334e5
stirrup	722	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5

Analysis and solver

analysis	pushover
target	2.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	4.416 us, 4,556 steps per fundamental period
first period T_1	20.12 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	2,726,278
wall clock	13.32 h
converged	True

Recorded response

peak base shear	139.4 kN (5 ms moving average)
raw sample maximum	162.3 kN
drift at peak	0.497%
drift capacity (20% drop)	2.374% — basis: monotonic
traced to	2.5% drift
inertia + damping in the reaction, ascending branch (p95)	1.3 kN

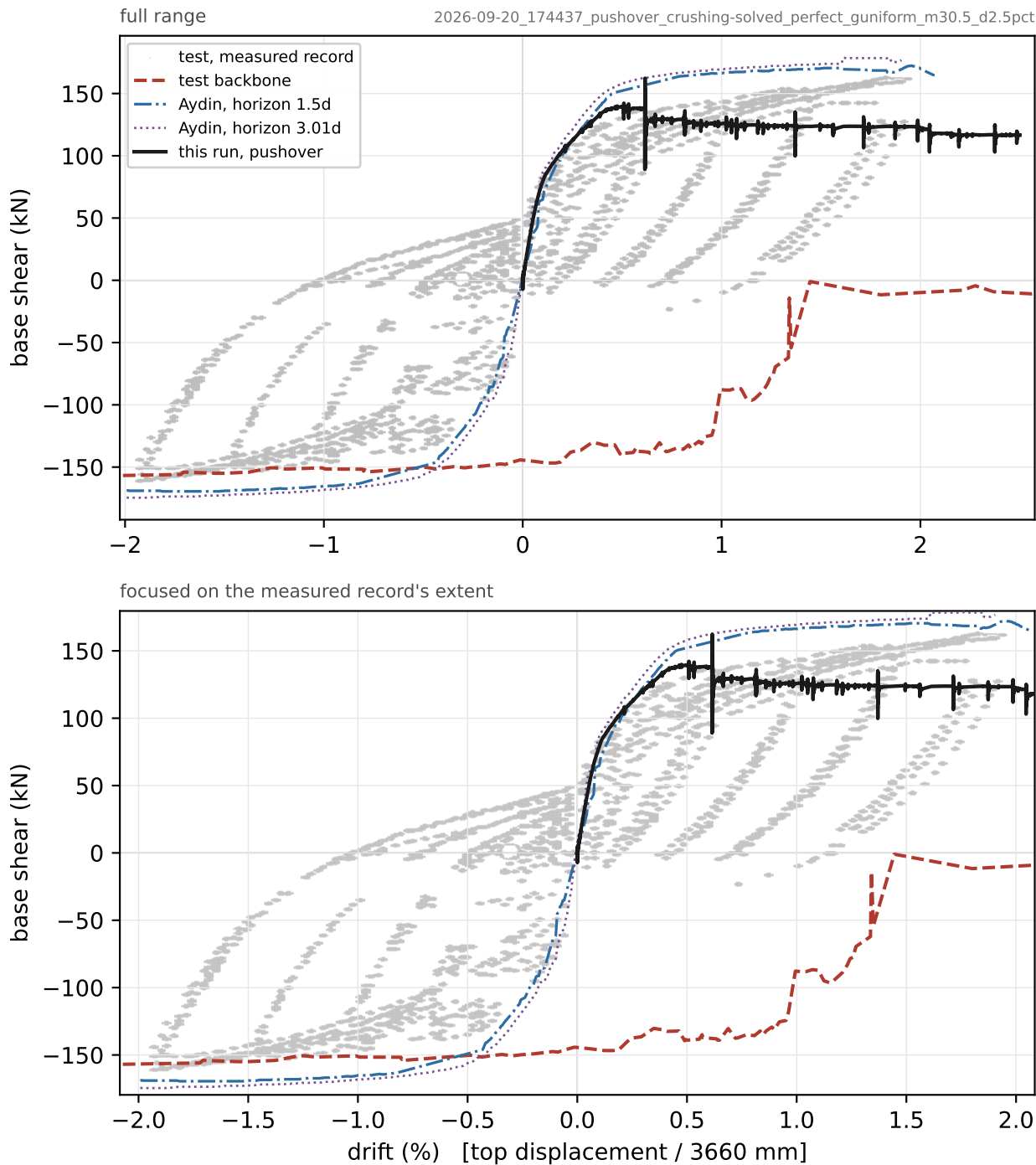


Figure 6: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

3 linear compression — solved tension tail — perfect bond

3.1 Pushover — linear/solved/perfect — mesh 30.5 mm — energy balance, uniaxial field — uniform grid — $f_y = 454$ — to 1.5% drift

2026-09-22_013850_pushover_linear-solved_perfect_guniform_m30.5_fy454_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	30.5 mm
strut horizon	$1.5 \times$ mesh
nodes / elements	4,961 / 22,002
composition	4,961 nodes, 22,002 elements (19,360 concrete, 1,440 longitudinal, 1,202 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 38, diagonal 27.2
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,889.1 \text{ mm}^2$
$E_t A_t$ per strut	58.6e6 N
G_f multiple of the tail	1.03 (1.03 x G_f per cracking strut at $L = 30.5$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8$ MPa, $f_t = 2.03$ MPa, $G_f = 0.075$ N/mm
compression branch	linear elastic in compression, never yields
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$

steel	$f_y = 414$ MPa, $E_s = 200,000$ MPa, $b = 0.01$
bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>materials</code> , <code>steel_iso</code> , <code>top_band</code> , <code>top_band_pitch</code> — this run predates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00091954, 0, 6.54206e-05, 9.81308e-05, 0.00166023, 0.00726352, 0.2], -stress: [-2499.52, -28.5333, 0, 2.03, 1.218, 0.406, 0, 0]	9,760 concrete elements, $L = 30.5$ mm
2	ElasticMultiLinear	-strain:[-0.2, -0.00091954, 0, 6.54206e-05, 9.81308e-05, 0.00117916, 0.00515882, 0.2], -stress: [-2499.52, -28.5333, 0, 2.03, 1.218, 0.406, 0, 0]	9,600 concrete elements, $L = 43.13$ mm
3	Steel02	Fy:454, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	2,642 longitudinal/stirrup elements, $L = 30.5$ mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	9,760	corotTruss	30.5	1,889.1	31,030	5.862e7	1.922e6
concrete, diagonal	9,600	corotTruss	43.13	1,889.1	31,030	5.862e7	1.359e6
longitudinal	960	corotTruss	30.5	142.7	200,000	2.853e7	9.355e5
longitudinal	480	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	480	corotTruss	30.5	35.6	200,000	7.118e6	2.334e5
stirrup	722	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 4,556 steps per fundamental period
first period T_1	19.874 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,656,014
wall clock	6.77 h
converged	True

Recorded response

peak base shear	146.7 kN (5 ms moving average)
raw sample maximum	170.6 kN
drift at peak	0.588%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction,	2.4 kN
ascending branch (p95)	

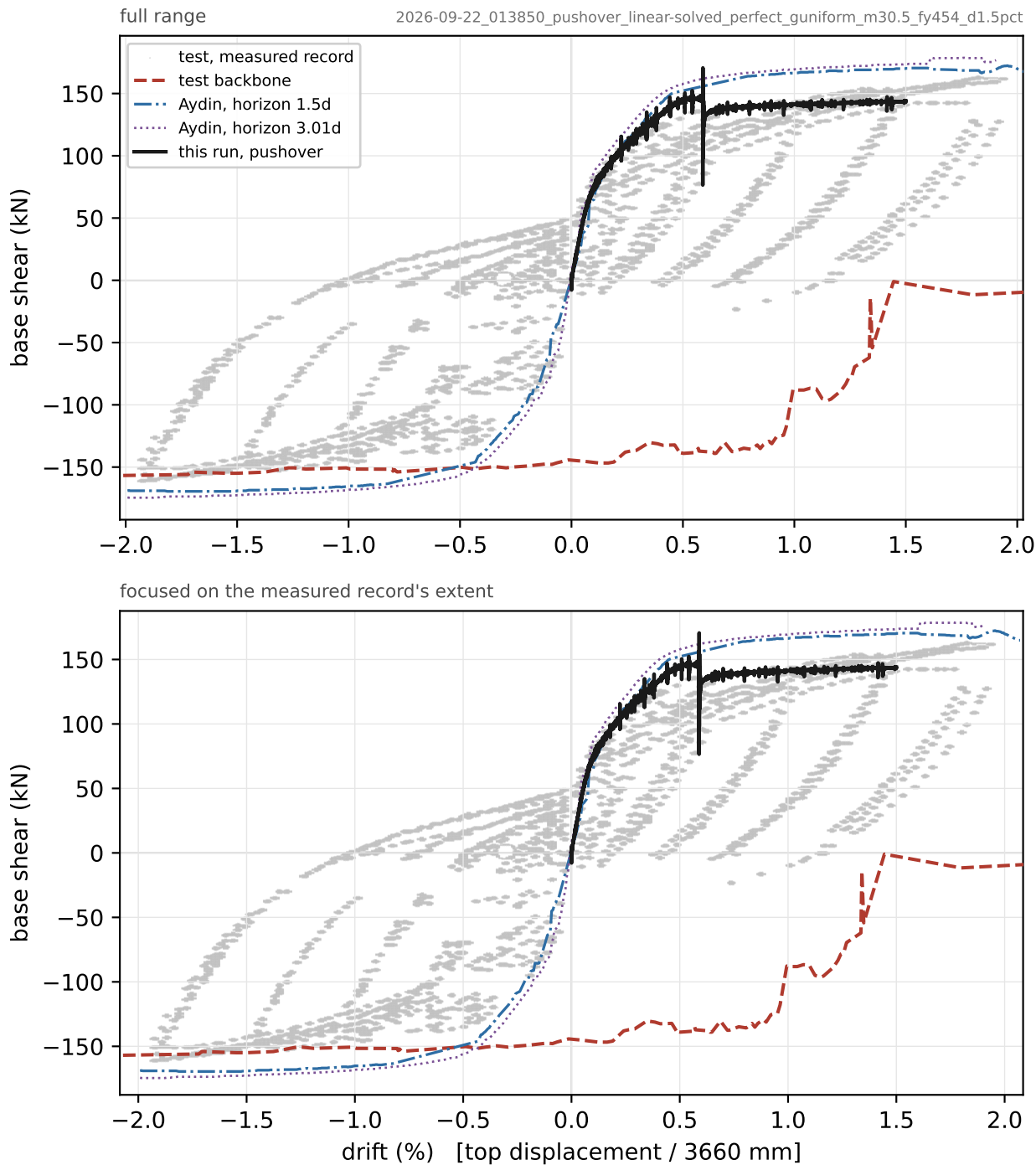


Figure 7: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

3.2 Pushover — linear/solved/perfect — mesh 30.5 mm — energy balance, uniaxial field — uniform grid — top band 1000 mm — to 2% drift

2026-09-22_164402_pushover_linear-solved_perfect_guniform_m30.5_tb1000_d2pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	30.5 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	4,961 / 22,310
composition	4,961 nodes, 22,310 elements (19,360 concrete, 1,440 longitudinal, 1,510 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 38, diagonal 27.2
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,889.1 \text{ mm}^2$
$E_t A_t$ per strut	58.6 e6 N
G_f multiple of the tail	1.03 (1.03 x Gf per cracking strut at $L = 30.5$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8 \text{ MPa}$, $f_t = 2.03 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	linear elastic in compression, never yields
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	none — bars cannot break

bond perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used

parameters filled from today's defaults materials, steel_iso — this run predates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00091954, 0, 6.54206e-05, 9.81308e-05, 0.00166023, 0.00726352, 0.2], -stress:[-2499.52, -28.5333, 0, 2.03, 1.218, 0.406, 0, 0]	9,760 concrete elements, L = 30.5 mm
2	ElasticMultiLinear	-strain:[-0.2, -0.00091954, 0, 6.54206e-05, 9.81308e-05, 0.00117916, 0.00515882, 0.2], -stress:[-2499.52, -28.5333, 0, 2.03, 1.218, 0.406, 0, 0]	9,600 concrete elements, L = 43.13 mm
3	Steel02	Fy:414, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	2,950 longitudinal/stirrup elements, L = 30.5 mm

Strut families — axial stiffness

note E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	9,760	corotTruss	30.5	1,889.1	31,030	5.862e7	1.922e6
concrete, diagonal	9,600	corotTruss	43.13	1,889.1	31,030	5.862e7	1.359e6
longitudinal	960	corotTruss	30.5	142.7	200,000	2.853e7	9.355e5
longitudinal	480	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5
stirrup	480	corotTruss	30.5	35.6	200,000	7.118e6	2.334e5
stirrup	1,030	corotTruss	30.5	63.3	200,000	1.267e7	4.153e5

Analysis and solver

analysis pushover
 target 2% drift
 solution scheme dynamic relaxation, quasi-static
 integrator CentralDifference
 time step 0 us, 4,556 steps per fundamental period
 first period T_1 19.874 ms
 drive rate 7.6 mm/s
 damping 0.5 critical, mass proportional
 steps 2,208,032
 wall clock 9 h
 converged True

Recorded response

peak base shear	139.9 kN (5 ms moving average)
raw sample maximum	168 kN
drift at peak	0.559%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	2% drift
inertia + damping in the reaction, ascending branch (p95)	2.3 kN

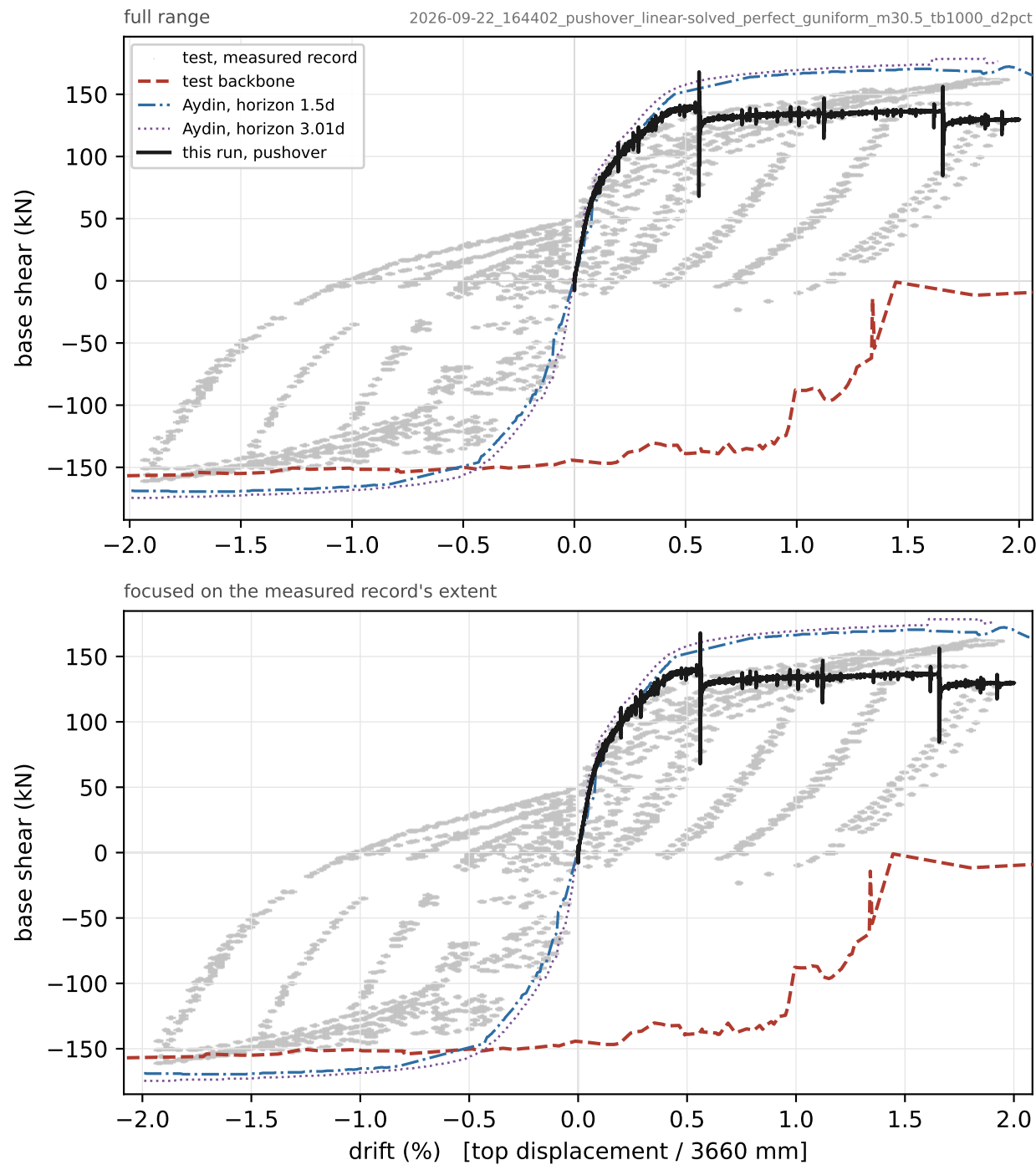


Figure 8: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

3.3 Pushover — linear/solved/perfect — mesh 30.5 mm — energy balance, uniaxial field — uniform grid — $b = 0.015$ — top band 1000 mm — to 2% drift

2026-09-23_080145_pushover_linear-solved_perfect_guniform_m30.5_sb0.015_tb1000_d2pct

Specimen and discretisation

panel ($L \times H \times t$)	$1,220 \times 3,660 \times 102$ mm
drift denominator	3,660 mm
mesh size	30.5 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	4,961 / 22,310
composition	4,961 nodes, 22,310 elements (19,360 concrete, 1,440 longitudinal, 1,510 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 38, diagonal 27.2
first cracking	35 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$1,889.1 \text{ mm}^2$
$E_t A_t$ per strut	58.6 e6 N
G_f multiple of the tail	1.03 (1.03 x Gf per cracking strut at $L = 30.5$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 42.8 \text{ MPa}$, $f_t = 2.03 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	linear elastic in compression, never yields
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 414 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.015$
bar rupture	none — bars cannot break

bond perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used

parameters filled from today's defaults materials, steel_iso — this run predates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00091954, 0, 6.54206e-05, 9.81308e-05, 0.00166023, 0.00726352, 0.2], -stress:[-2499.52, -28.5333, 0, 2.03, 1.218, 0.406, 0, 0]	9,760 concrete elements, L = 30.5 mm
2	ElasticMultiLinear	-strain:[-0.2, -0.00091954, 0, 6.54206e-05, 9.81308e-05, 0.00117916, 0.00515882, 0.2], -stress:[-2499.52, -28.5333, 0, 2.03, 1.218, 0.406, 0, 0]	9,600 concrete elements, L = 43.13 mm
3	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
4	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
5	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
6	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
7	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
8	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
9	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
10	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
11	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
12	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
13	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
14	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	120 longitudinal elements, L = 30.5 mm
15	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
16	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm
17	Steel02	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	5 stirrup elements, L = 30.5 mm

[illegible]

[illegible]

[illegible]

[illegible]

tag	type	arguments	applied to
134	Steel102	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	28 stirrup elements, L = 30.5 mm
135	Steel102	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	28 stirrup elements, L = 30.5 mm
136	Steel102	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	28 stirrup elements, L = 30.5 mm
137	Steel102	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	28 stirrup elements, L = 30.5 mm
138	Steel102	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	28 stirrup elements, L = 30.5 mm
139	Steel102	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	28 stirrup elements, L = 30.5 mm
140	Steel102	Fy:414, E0:200000, b:0.015, R0:18, cR1:0.925, cR2:0.15, a1:0.02, a2:1, a3:0.02, a4:1	28 stirrup elements, L = 30.5 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\epsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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[illegible]

[illegible]

[illegible]

Analysis and solver

analysis	pushover
target	2% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 4,556 steps per fundamental period
first period T_1	19.874 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	2,208,032
wall clock	9.61 h
converged	True

Recorded response

peak base shear	140 kN (5 ms moving average)
raw sample maximum	152 kN
drift at peak	1.995%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	2% drift
inertia + damping in the reaction, ascending branch (p95)	2.4 kN

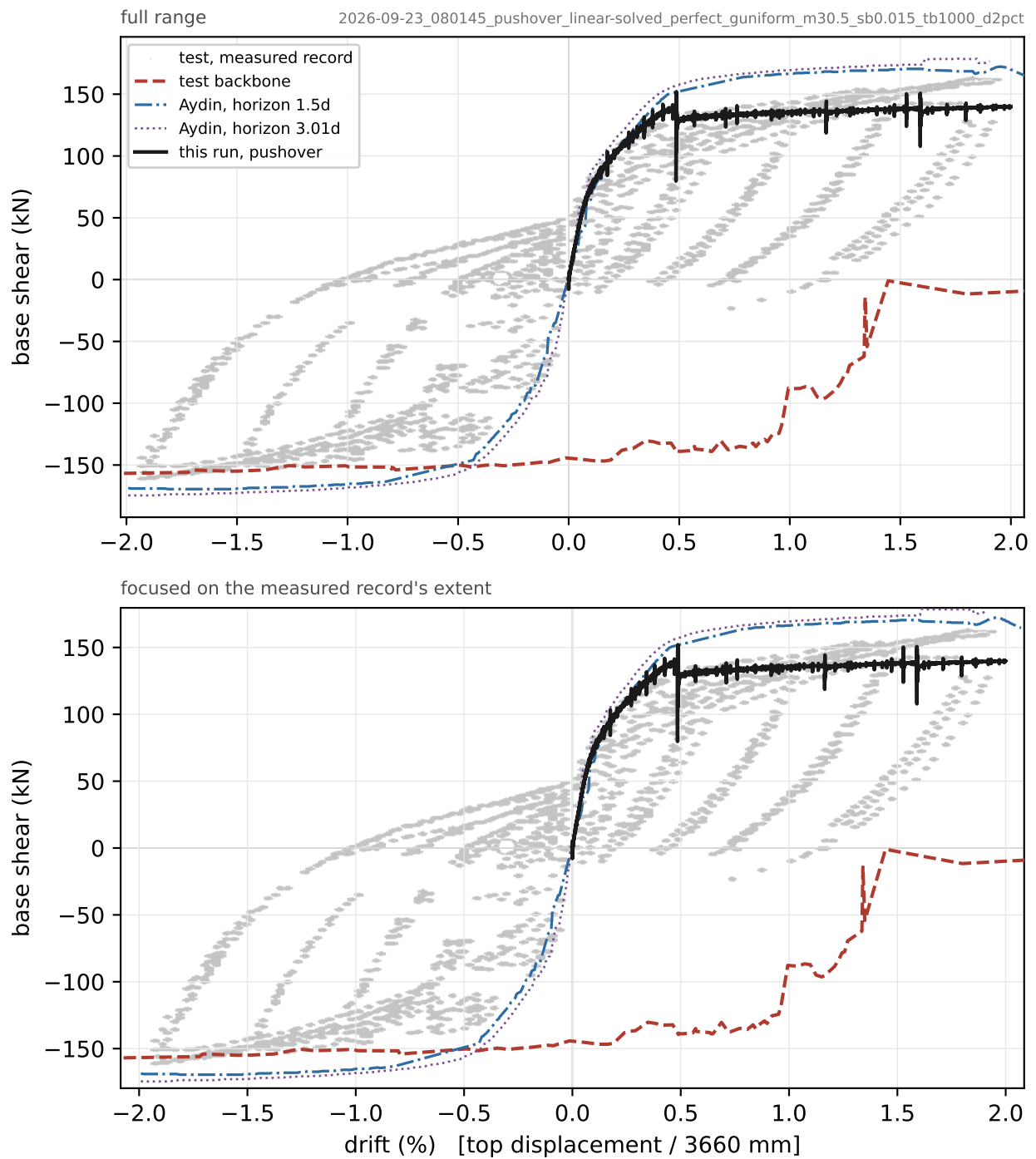


Figure 9: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.